

Infosys SP Placement — DSA Master Roadmap

Goal

Prepare the complete set of high-value DSA patterns reported/recommended for Infosys SP/HackWithInfy-style online assessments, in a **pattern-first order**. Each problem will later be taught using exactly:

1. Problem Statement
2. What is being asked
3. Sample Test Case(s)
4. Java Implementation Solution

The problems are ordered so that each pattern builds on the previous one.

Phase 1 — Core Array Foundations

Pattern 1: Array Traversal & Basic Optimization

1. Find Maximum and Minimum Element
2. Second Largest Element
3. Remove Duplicates from Sorted Array
4. Move Zeroes
5. Majority Element

Pattern 2: Prefix Sum & Pre-computation

6. Prefix Sum — Range Sum Queries
7. Prefix/Suffix Sum — Best Split Point
8. Maximum Subarray Sum — Kadane's Algorithm
9. Range Sum Query with Multiple Queries

- 10. Prefix Frequency — Repeated Frequency Queries
 - 11. Range Property Queries Using Pre-computation
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Phase 2 — Hashing, Frequency Maps & Subarrays

Pattern 3: HashMap / HashSet

- 12. Two Sum
- 13. Longest Consecutive Sequence
- 14. Count Frequencies of Array Elements
- 15. Subarray Sum Equals K
- 16. Longest Subarray with Sum K
- 17. Count Subarrays with Given Sum

Pattern 4: Sliding Window + HashMap

- 18. Longest Substring Without Repeating Characters
 - 19. Longest Subarray with At Most K Distinct Elements
 - 20. Count Distinct Elements in Every Window
 - 21. Minimum Window Substring
 - 22. Constrained Subarray Using Hashing + Sliding Window
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Phase 3 — Two Pointers & Sorting-Based Patterns

Pattern 5: Two Pointers

- 23. Two Sum in Sorted Array
- 24. 3Sum
- 25. Container With Most Water
- 26. Remove Duplicates Using Two Pointers
- 27. Merge Two Sorted Arrays

Pattern 6: Sorting + Greedy Array Manipulation

- 28. Minimum Operations to Make Array Ordered
 - 29. Minimum Cost Array Transformation
 - 30. Maximum Sum After Selecting Elements with a Distance Constraint
 - 31. Partition Array to Maximize Score
 - 32. Advanced Array Manipulation and Optimization
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Phase 4 — Greedy Algorithms

Pattern 7: Interval Greedy

- 33. Activity Selection
- 34. Greedy Interval Scheduling — Maximum Non-overlapping Tasks
- 35. Merge Intervals
- 36. Minimum Number of Platforms
- 37. Job Sequencing with Deadlines

Pattern 8: Greedy Optimization

- 38. Fractional Knapsack
- 39. Minimum Number of Operations Using Greedy Choices
- 40. Minimum Cost Transformation
- 41. Maximum Value Under Selection Constraints

Pattern 9: Binary Search on Answer

- 42. Koko Eating Bananas
 - 43. Capacity to Ship Packages Within D Days
 - 44. Aggressive Cows
 - 45. Allocate Books
 - 46. Minimum Time / Maximum Feasible Answer Problems
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Phase 5 — Dynamic Programming Fundamentals

Pattern 10: 1D DP

- 47. Climbing Stairs
- 48. House Robber
- 49. Frog Jump
- 50. Maximum Sum of Non-adjacent Elements

Pattern 11: Knapsack / Selection DP

- 51. 0/1 Knapsack
- 52. Subset Sum
- 53. Partition Equal Subset Sum
- 54. Count Ways to Reach a Target Sum
- 55. Coin Change
- 56. Target Sum

Pattern 12: String DP

- 57. Longest Common Subsequence
- 58. Longest Common Substring
- 59. Edit Distance
- 60. Minimum Cost to Transform One String into Another
- 61. Distinct Subsequences

Pattern 13: Increasing Sequence DP

- 62. Longest Increasing Subsequence — $O(N^2)$
- 63. LIS — $O(N \log N)$
- 64. Longest Divisible Subset
- 65. Longest Increasing Subsequence with Difference Constraint

Pattern 14: DP Optimization

- 66. Optimize $O(N^2)$ DP to $O(N \log N)$

- 67. Space Optimization in DP
 - 68. Tabulation Optimization to Avoid TLE
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Phase 6 — Trees

Pattern 15: Tree Traversals

- 69. Binary Tree Preorder Traversal
- 70. Binary Tree Inorder Traversal
- 71. Binary Tree Postorder Traversal
- 72. Level Order Traversal
- 73. Iterative DFS/BFS Traversals

Pattern 16: Tree Properties

- 74. Height / Maximum Depth of Binary Tree
- 75. Diameter of Binary Tree
- 76. Diameter of a Weighted Tree
- 77. Maximum Path Sum in a Binary Tree

Pattern 17: Lowest Common Ancestor

- 78. LCA of Two Nodes in Binary Tree
- 79. LCA in Binary Search Tree
- 80. LCA Using Binary Lifting

Pattern 18: Subtree / Tree DP

- 81. Subtree Sum
- 82. Subtree Queries
- 83. Maximum Independent Set / Selection on Tree
- 84. Minimum Operations on a Tree
- 85. Tree DP

Pattern 19: Rerooting

- 86. Sum of Distances from Every Node
 - 87. DFS Rerooting — Calculate a Value for Every Node
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Phase 7 — Graph Fundamentals

Pattern 20: Graph Representation & Traversal

- 88. Graph Representation Using Adjacency List
- 89. BFS of a Graph
- 90. DFS of a Graph
- 91. Connected Components

Pattern 21: Cycle Detection

- 92. Cycle Detection in Undirected Graph — BFS
- 93. Cycle Detection in Undirected Graph — DFS
- 94. Cycle Detection in Directed Graph — DFS
- 95. Cycle Detection Using Kahn's Algorithm

Pattern 22: Bipartite Graph

- 96. Bipartite Graph Using BFS
- 97. Bipartite Graph Using DFS

Pattern 23: Topological Sorting

- 98. Topological Sort Using DFS
- 99. Topological Sort Using Kahn's Algorithm
- 100. Course Schedule / Dependency Ordering

Phase 8 — Shortest Path Algorithms

Pattern 24: Unweighted Shortest Path

- 101. Shortest Path in an Unweighted Graph
- 102. Multi-source BFS

Pattern 25: Dijkstra

- 103. Dijkstra's Shortest Path
- 104. Shortest Path in Weighted Graph
- 105. Network Delay / Minimum Travel Time

Pattern 26: Other Shortest Path Patterns

- 106. Bellman-Ford
 - 107. Floyd-Warshall
 - 108. Shortest Path in DAG
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Phase 9 — Minimum Spanning Tree & DSU

Pattern 27: Minimum Spanning Tree

- 109. Minimum Spanning Tree — Kruskal
- 110. Minimum Spanning Tree — Prim
- 111. Minimum Cost to Connect All Cities

Pattern 28: DSU / Union-Find

- 112. Disjoint Set Union Basics
 - 113. Number of Connected Components Using DSU
 - 114. Cycle Detection Using DSU
 - 115. Connectivity Queries
 - 116. Dynamic Connectivity After Edge Updates
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Phase 10 — Advanced Data Structures

Pattern 29: Fenwick Tree / BIT

- 117. Fenwick Tree — Point Update + Prefix Query
- 118. Range Sum Queries Using BIT
- 119. Dynamic Frequency Queries Using BIT

Pattern 30: Segment Tree

- 120. Segment Tree — Range Sum Query
- 121. Segment Tree — Range Minimum Query
- 122. Segment Tree — Range Maximum Query
- 123. Segment Tree — Point Update + Range Query
- 124. Segment Tree — Range Update + Range Query

Pattern 31: Advanced Range Queries

- 125. Dynamic Range Minimum/Maximum Queries
- 126. Range Updates with Lazy Propagation
- 127. Multiple Dynamic Range Queries

Pattern 32: Order Statistics / Frequency

- 128. Frequency Query Using Fenwick Tree
 - 129. Order-Statistic Style Queries
 - 130. Coordinate Compression + BIT
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Phase 11 — Integrated Advanced Patterns

Pattern 33: Hashing + Sliding Window

- 131. Constrained Subarray Using HashMap
- 132. Longest Valid Subarray
- 133. Frequency-Constrained Window

Pattern 34: Array + Binary Search + Greedy

- 134. Minimum Feasible Value
- 135. Maximum Achievable Value
- 136. Binary Search on Answer with Greedy Validation

Pattern 35: DP + Binary Search

- 137. LIS with Binary Search
- 138. Optimized DP with Binary Search
- 139. DP State Optimization

Pattern 36: Tree + Graph Integration

- 140. Tree as an Undirected Graph
- 141. Tree Diameter Using Two BFS/DFS
- 142. Tree Queries Using DFS + Pre-computation
- 143. LCA + Binary Lifting
- 144. Rerooting DP

Phase 12 — Infosys SP-Level Mixed Problems

Pattern 37: Easy-Level Assessment Patterns

- 145. Greedy + Arrays Selection Problem
- 146. Prefix Sum / Pre-computation Problem
- 147. HashMap / Frequency Query Problem
- 148. Sliding Window Problem

Pattern 38: Medium-Level Assessment Patterns

- 149. Advanced Array Manipulation
- 150. DP with Optimized Tabulation
- 151. DP with $O(N \log N)$ Optimization
- 152. Binary Search on Answer
- 153. Segment Tree / Fenwick Tree Problem

Pattern 39: Hard-Level Assessment Patterns

- 154. Hard Tree Algorithm
- 155. Tree DP / Rerooting Problem
- 156. Hard Graph Optimization Problem
- 157. Advanced Data Structure Problem
- 158. Segment Tree with Lazy Propagation
- 159. DSU + Offline/Online Connectivity Problem

Phase 13 — Final Must-Master Set

These are the highest-priority problems to revise repeatedly:

- 160. Maximum Subarray Sum — Kadane's Algorithm
- 161. Longest Common Subsequence
- 162. Longest Increasing Subsequence
- 163. Edit Distance
- 164. 0/1 Knapsack
- 165. Coin Change
- 166. Sliding Window + HashMap
- 167. Prefix Sum + Range Queries
- 168. Greedy Interval Scheduling
- 169. Binary Search on Answer
- 170. Tree Diameter
- 171. Lowest Common Ancestor
- 172. Subtree Sum / Tree DP

173. Tree Rerooting
174. Cycle Detection
175. Bipartite Graph
176. Topological Sort
177. Dijkstra Shortest Path
178. Minimum Spanning Tree — Kruskal + DSU
179. Minimum Spanning Tree — Prim
180. DSU / Union-Find Connectivity
181. Fenwick Tree / BIT
182. Segment Tree
183. Segment Tree + Lazy Propagation
184. Hashing + Frequency Maps
185. Advanced Array Optimization
186. DP Tabulation Optimization
187. $O(N^2) \rightarrow O(N \log N)$ Optimization

Recommended Learning Order

Level 1 — Must Finish First

Arrays $\rightarrow$ Prefix Sum $\rightarrow$ HashMap/HashSet $\rightarrow$ Two Pointers $\rightarrow$ Sliding Window

Level 2 — Optimization

Sorting $\rightarrow$ Greedy $\rightarrow$ Binary Search $\rightarrow$ Binary Search on Answer

Level 3 — Dynamic Programming

1D DP $\rightarrow$ Knapsack $\rightarrow$ String DP $\rightarrow$ LIS $\rightarrow$ DP Optimization

Level 4 — Trees

Traversal $\rightarrow$ Tree Properties $\rightarrow$ Diameter $\rightarrow$ LCA $\rightarrow$ Subtree Queries $\rightarrow$ Tree DP $\rightarrow$ Rerooting

Level 5 — Graphs

BFS/DFS $\rightarrow$ Components $\rightarrow$ Cycle Detection $\rightarrow$ Bipartite $\rightarrow$ Topological Sort $\rightarrow$ Shortest Path

Level 6 — Advanced Graphs

Dijkstra → MST → Kruskal → Prim → DSU → Dynamic Connectivity

Level 7 — Advanced Data Structures

Fenwick Tree → Segment Tree → Lazy Propagation → Order Statistics → Coordinate Compression

Level 8 — Final Integration

Mixed Infosys SP Problems → Timed Practice → Previous/Reported Patterns → Full Mock Tests

Teaching Format for Every Problem

For every problem from this index, the notes will follow exactly this structure:

Problem X — Problem Title

1. Problem Statement

Clear interview/assessment-style statement.

2. What Is Being Asked?

The exact objective and what the output represents.

3. Sample Test Case

Input and expected output with a short explanation.

4. Java Implementation Solution

Clean, placement-ready Java code with the relevant pattern.

Priority

Highest Priority: 160–187

Next Priority: 47–68, 69–87, 88–116, 117–139

Foundation: 1–46

The teaching should proceed sequentially. After completing one problem, continue with the next problem in the index so that the patterns build naturally from basic to Infosys SP-level advanced problems.